

CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 7: Coordinate Geometry

Step-by-step solutions to the textbook exercises: 20 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 7.1

Q1 Find the distance between each pair of points: (i) (2, 3), (4, 1) (ii) (-5, 7), (-1, 3) (iii) (a, b), (-a, -b).

SOLUTION

Use the distance formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

(i) $\sqrt{(4 - 2)^2 + (1 - 3)^2} = \sqrt{4 + 4} = 2\sqrt{2}$

(ii) $\sqrt{(-1 + 5)^2 + (3 - 7)^2} = \sqrt{16 + 16} = 4\sqrt{2}$

(iii) $\sqrt{(-2a)^2 + (-2b)^2} = \sqrt{4a^2 + 4b^2} = 2\sqrt{a^2 + b^2}$

Answer: (i) $2\sqrt{2}$ (ii) $4\sqrt{2}$ (iii) $2\sqrt{a^2 + b^2}$ units

Q2 Find the distance between (0, 0) and (36, 15). Hence find the distance between the two towns of Section 7.2 of the textbook, where town B is 36 km east and 15 km north of town A.

SOLUTION

Distance = $\sqrt{36^2 + 15^2} = \sqrt{1296 + 225} = \sqrt{1521} = 39$.

Taking town A as the origin, town B is the point (36, 15), so the towns are 39 km apart.

Answer: 39 units, so the towns are 39 km apart

Q3 Are the points (1, 5), (2, 3) and (-2, -11) collinear?

SOLUTION

Call them A, B and C. $AB = \sqrt{1 + 4} = \sqrt{5} \approx 2.24$, $BC = \sqrt{16 + 196} = \sqrt{212} \approx 14.56$ and $AC = \sqrt{9 + 256} = \sqrt{265} \approx 16.28$.

The two shorter distances add up to about 16.80, which is not equal to the longest distance 16.28.

Answer: no, the points are not collinear

Q4 Are (5, -2), (6, 4) and (7, -2) the vertices of an isosceles triangle?

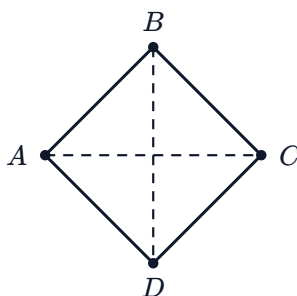
SOLUTION

Call them A, B and C. $AB = \sqrt{1 + 36} = \sqrt{37}$, $BC = \sqrt{1 + 36} = \sqrt{37}$ and $AC = \sqrt{4 + 0} = 2$.

$AB = BC$, and the points are not collinear since $AB + AC \neq BC$.

Answer: yes, the triangle is isosceles with $AB = BC$

- Q5** Four friends sit at the points $A(3, 4)$, $B(6, 7)$, $C(9, 4)$ and $D(6, 1)$ of a classroom grid (read from the textbook figure). One student says $ABCD$ is a square and another disagrees. Who is correct?



$A(3, 4)$, $B(6, 7)$, $C(9, 4)$, $D(6, 1)$

SOLUTION

Sides: $AB = \sqrt{3^2 + 3^2} = \sqrt{18}$, $BC = \sqrt{3^2 + 3^2} = \sqrt{18}$, $CD = \sqrt{18}$ and $DA = \sqrt{18}$. All four sides are equal.

Diagonals: $AC = \sqrt{6^2 + 0^2} = 6$ and $BD = \sqrt{0^2 + 6^2} = 6$. The diagonals are equal.

A quadrilateral with four equal sides and equal diagonals is a square.

Answer: $ABCD$ is a square, so the student who said so is correct

- Q6** Name the type of quadrilateral formed, if any, by these points, with reasons: (i) $(-1, -2)$, $(1, 0)$, $(-1, 2)$, $(-3, 0)$ (ii) $(-3, 5)$, $(3, 1)$, $(0, 3)$, $(-1, -4)$ (iii) $(4, 5)$, $(7, 6)$, $(4, 3)$, $(1, 2)$.

SOLUTION

Call the points A, B, C, D in order.

(i) $AB = BC = CD = DA = \sqrt{4 + 4} = \sqrt{8}$. Diagonals: $AC = 4$ and $BD = 4$. All sides are equal and the diagonals are equal, so it is a square.

(ii) $AB = \sqrt{36 + 16} = 2\sqrt{13}$, $BC = \sqrt{9 + 4} = \sqrt{13}$ and $AC = \sqrt{9 + 4} = \sqrt{13}$. Since $AC + CB = AB$, the points A, C, B lie on one line, so no quadrilateral is formed.

(iii) $AB = \sqrt{10}$, $BC = \sqrt{18}$, $CD = \sqrt{10}$, $DA = \sqrt{18}$. Diagonals: $AC = 2$ and $BD = \sqrt{36 + 16} = \sqrt{52}$. Opposite sides are equal but the diagonals are not, so it is a parallelogram.

Answer: (i) a square (ii) no quadrilateral (iii) a parallelogram

- Q7** Find the point on the x -axis that is equidistant from $(2, -5)$ and $(-2, 9)$.

SOLUTION

Let the point be $(x, 0)$. Then $(x - 2)^2 + 25 = (x + 2)^2 + 81$.

$x^2 - 4x + 29 = x^2 + 4x + 85$ gives $-8x = 56$, so $x = -7$.

Answer: $(-7, 0)$

Q8 Find the values of y for which the distance between $P(2, -3)$ and $Q(10, y)$ is 10 units.

SOLUTION

$$(10 - 2)^2 + (y + 3)^2 = 100 \text{ gives } (y + 3)^2 = 36, \text{ so } y + 3 = \pm 6.$$

Answer: $y = 3$ or $y = -9$

Q9 $Q(0, 1)$ is equidistant from $P(5, -3)$ and $R(x, 6)$. Find the values of x , and the distances QR and PR .

SOLUTION

$$QP^2 = 25 + 16 = 41 \text{ and } QR^2 = x^2 + 25. \text{ Setting them equal, } x^2 = 16, \text{ so } x = \pm 4.$$

In both cases $QR = \sqrt{41}$.

$$\text{If } R = (4, 6): PR = \sqrt{1 + 81} = \sqrt{82}. \text{ If } R = (-4, 6): PR = \sqrt{81 + 81} = 9\sqrt{2}.$$

Answer: $x = \pm 4$, $QR = \sqrt{41}$, and $PR = \sqrt{82}$ or $9\sqrt{2}$

Q10 Find a relation between x and y such that (x, y) is equidistant from $(3, 6)$ and $(-3, 4)$.

SOLUTION

$$(x - 3)^2 + (y - 6)^2 = (x + 3)^2 + (y - 4)^2.$$

$$-6x - 12y + 45 = 6x - 8y + 25 \text{ gives } 12x + 4y - 20 = 0.$$

Answer: $3x + y - 5 = 0$

2 Exercise 7.2

Q1 Find the point that divides the join of $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$.

SOLUTION

$$\text{Section formula: } \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right) = \left(\frac{2 \times 4 + 3 \times (-1)}{5}, \frac{2 \times (-3) + 3 \times 7}{5} \right).$$

Answer: $(1, 3)$

Q2 Find the points of trisection of the segment joining $(4, -1)$ and $(-2, -3)$.

SOLUTION

Let $A = (4, -1)$ and $B = (-2, -3)$. The first point P divides AB in the ratio $1 : 2$: $P =$

$$\left(\frac{-2 + 8}{3}, \frac{-3 - 2}{3} \right) = \left(2, -\frac{5}{3} \right).$$

The second point Q divides AB in the ratio $2 : 1$: $Q = \left(\frac{-4 + 4}{3}, \frac{-6 - 1}{3} \right) = \left(0, -\frac{7}{3} \right).$

Answer: $\left(2, -\frac{5}{3} \right)$ and $\left(0, -\frac{7}{3} \right)$

Q3 On a rectangular ground $ABCD$, lines are drawn 1 m apart, and 100 flower pots stand 1 m apart along AD . One student runs $\frac{1}{4}$ of AD along the 2nd line and posts a green flag. Another runs $\frac{1}{5}$ of AD along the 8th line and posts a red flag. Find the distance between the flags. Where should a blue flag go so that it is exactly halfway between them?

SOLUTION

Take A as the origin, AB as the x -axis and AD as the y -axis, with $AD = 100$ m.

Green flag: on the 2nd line at $\frac{1}{4} \times 100 = 25$ m, that is $(2, 25)$. Red flag: on the 8th line at $\frac{1}{5} \times 100 = 20$ m, that is $(8, 20)$.

Distance = $\sqrt{(8-2)^2 + (20-25)^2} = \sqrt{61}$ m.

Mid-point = $\left(\frac{2+8}{2}, \frac{25+20}{2}\right) = (5, 22.5)$.

Answer: $\sqrt{61}$ m apart. The blue flag goes on the 5th line, 22.5 m from AB .

Q4 In what ratio does $(-1, 6)$ divide the segment joining $(-3, 10)$ and $(6, -8)$?

SOLUTION

Let the ratio be $k : 1$. Then $\frac{6k-3}{k+1} = -1$ gives $7k = 2$, so $k = \frac{2}{7}$.

Check with y : $\frac{-8 \times 2 + 10 \times 7}{9} = 6$.

Answer: $2 : 7$

Q5 In what ratio does the x -axis divide the segment joining $A(1, -5)$ and $B(-4, 5)$? Find the point of division.

SOLUTION

Let the ratio be $k : 1$. The y -coordinate of the point is 0: $\frac{5k-5}{k+1} = 0$, so $k = 1$.

The point is the mid-point: $\left(\frac{1-4}{2}, 0\right) = \left(-\frac{3}{2}, 0\right)$.

Answer: $1 : 1$, at $\left(-\frac{3}{2}, 0\right)$

Q6 $(1, 2)$, $(4, y)$, $(x, 6)$ and $(3, 5)$ are the vertices of a parallelogram taken in order. Find x and y .

SOLUTION

The diagonals of a parallelogram bisect each other, so the two diagonals have the same mid-point.

$\frac{1+x}{2} = \frac{4+3}{2}$ gives $x = 6$, and $\frac{2+y}{2} = \frac{5+6}{2}$ gives $y = 3$.

Answer: $x = 6, y = 3$

Q7 AB is a diameter of a circle with centre $(2, -3)$, and B is $(1, 4)$. Find A .

SOLUTION

The centre is the mid-point of AB . Let $A = (x, y)$: $\frac{x+1}{2} = 2$ gives $x = 3$, and $\frac{y+4}{2} = -3$ gives $y = -10$.

Answer: $A = (3, -10)$

Q8 A and B are $(-2, -2)$ and $(2, -4)$. Find the point P on the segment AB such that $AP = \frac{3}{7}AB$.

SOLUTION

$AP = \frac{3}{7}AB$ means $PB = \frac{4}{7}AB$, so P divides AB in the ratio $3 : 4$.

$$P = \left(\frac{3 \times 2 + 4 \times (-2)}{7}, \frac{3 \times (-4) + 4 \times (-2)}{7} \right) = \left(-\frac{2}{7}, -\frac{20}{7} \right).$$

Answer: $P = \left(-\frac{2}{7}, -\frac{20}{7} \right)$

Q9 Find the points that divide the segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.

SOLUTION

The middle point is the mid-point of AB : $(0, 5)$.

The other two are the mid-points of each half: $\left(\frac{-2+0}{2}, \frac{2+5}{2} \right) = \left(-1, \frac{7}{2} \right)$ and $\left(\frac{0+2}{2}, \frac{5+8}{2} \right) = \left(1, \frac{13}{2} \right)$.

Answer: $\left(-1, \frac{7}{2} \right)$, $(0, 5)$ and $\left(1, \frac{13}{2} \right)$

Q10 Find the area of the rhombus with vertices $(3, 0)$, $(4, 5)$, $(-1, 4)$ and $(-2, -1)$ taken in order.

SOLUTION

Call the vertices A, B, C, D . Diagonals: $AC = \sqrt{(-4)^2 + 4^2} = 4\sqrt{2}$ and $BD = \sqrt{(-6)^2 + (-6)^2} = 6\sqrt{2}$.

$$\text{Area} = \frac{1}{2} \times AC \times BD = \frac{1}{2} \times 4\sqrt{2} \times 6\sqrt{2} = 24.$$

Answer: 24 square units

The questions are restated in MathSetu's own words and the figures are redrawn; refer to the textbook for the original wording. The solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by NCERT, CBSE or any publisher.